常用公式记忆总结

第一章误差

绝对误差与相对误差

$$e = |x - x^*|, \qquad e_r = \frac{|x - x^*|}{|x|}$$

有效数字误差界

$$e_r < \frac{1}{2a_1} 10^{1-n}$$

幂函数误差传播

$$y = x^m, \qquad e_r(y) \approx |m| e_r(x)$$

$$S = \pi r^2, \qquad e_r(S) \approx 2e_r(r)$$

对数函数误差传播

$$y = \ln x, \qquad \Delta y \approx \frac{\Delta x}{x}$$

秦九韶格式

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = (\cdots((a_n x + a_{n-1})x + a_{n-2})x + \cdots)x + a_0$$

递推稳定性

$$e_n = c e_{n-1}$$

$$|c| < 1 \Rightarrow \text{稳定}, \qquad |c| > 1 \Rightarrow \text{不稳定}$$

第二章插值

Lagrange 插值

$$P_n(x) = \sum_{i=0}^{n} y_i l_i(x)$$

$$l_i(x) = \prod_{j=0, j\neq i}^{n} \frac{x - x_j}{x_i - x_j}$$

$$l_i(x_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j \end{cases}$$

插值余项

$$R_n(x) = f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i)$$

Newton 插值

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + \cdots + f[x_0, \dots, x_n] \prod_{k=0}^{n-1} (x - x_k)$$

差商

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

$$f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Hermite / 重节点

$$f(a),\ f'(a) \quad \Rightarrow \quad \text{Hermite 插值}$$

$$p(a) = 0,\ p'(a) = 0 \quad \Rightarrow \quad (x - a)^2 \mid p(x)$$

第三章最小二乘与平方逼近

离散最小二乘

$$\varphi(x) = \sum_{j=0}^{n} a_j \phi_j(x)$$

$$\sum_{j=0}^{n} a_j \sum_{k=1}^{m} \phi_j(x_k)\phi_i(x_k) = \sum_{k=1}^{m} y_k \phi_i(x_k), \qquad i = 0, 1, \dots, n$$

$$\boxed{A^T A a = A^T y}$$

带权最小二乘

$$\sum_{k=1}^{m} \omega_k \big(\varphi(x_k) - y_k\big)^2 \to \min$$

$$\sum_{j=0}^{n} a_j \sum_{k=1}^{m} \omega_k \phi_j(x_k)\phi_i(x_k) = \sum_{k=1}^{m} \omega_k y_k \phi_i(x_k)$$

常见拟合模型

$$y = a + bx: \quad \begin{cases} ma + b\sum x_i = \sum y_i, \\ a\sum x_i + b\sum x_i^2 = \sum x_i y_i \end{cases}$$

$$y = a + bx^2: \quad \begin{cases} ma + b\sum x_i^2 = \sum y_i, \\ a\sum x_i^2 + b\sum x_i^4 = \sum x_i^2 y_i \end{cases}$$

$$y = a + bx + cx^2: \quad \begin{cases} ma + b\sum x_i + c\sum x_i^2 = \sum y_i, \\ a\sum x_i + b\sum x_i^2 + c\sum x_i^3 = \sum x_i y_i, \\ a\sum x_i^2 + b\sum x_i^3 + c\sum x_i^4 = \sum x_i^2 y_i \end{cases}$$

$$y = ax + bx^2: \quad \begin{pmatrix} \sum x_i^2 & \sum x_i^3 \\ \sum x_i^3 & \sum x_i^4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \sum x_i y_i \\ \sum x_i^2 y_i \end{pmatrix}$$

连续最佳平方逼近

$$P_n(x) = \sum_{j=0}^{n} a_j \phi_j(x)$$

$$\int_a^b (f(x) - P_n(x))\phi_i(x)\,dx = 0, \qquad i = 0, 1, \dots, n$$

$[-a, a]$ 上偶函数的最佳逼近不含奇次项

二次型函数导数

$$F(x) = \|Ax - y\|_2^2 + \|x\|_2^2$$

$$\boxed{F'(x) = 2A^T(Ax - y) + 2x}$$

共轭梯度法

$$\min\left(\frac{1}{2}x^T A x - b^T x + c\right) \quad \Longleftrightarrow \quad Ax = b$$

$$r_k = b - Ax_k, \qquad p_0 = r_0$$

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}$$

$$x_{k+1} = x_k + \alpha_k p_k$$

$$r_{k+1} = r_k - \alpha_k A p_k$$

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}, \qquad p_{k+1} = r_{k+1} + \beta_k p_k$$

第四章数值积分与微分

Newton-Cotes 常用公式

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}[f(a) + f(b)]$$

$$\int_a^b f(x)\,dx \approx \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]$$

复化公式

$$T_n = h\left[\frac{1}{2}f_0+\sum_{i=1}^{n-1}f_i+\frac{1}{2}f_n\right]$$

$$S_n = \frac{h}{3}\left[f_0+f_n+4(f_1+f_3+\cdots+f_{n-1})+2(f_2+f_4+\cdots+f_{n-2})\right], \qquad n \text{ 为偶数}$$

代数精度

$$1, x, x^2, \ldots, x^m \text{ 均精确} \quad \Rightarrow \quad \text{代数精度为} m$$

Gauss 型求积

$$n+1 \text{ 个节点} \quad \Rightarrow \quad \text{最高代数精度} 2n+1$$

四点前向差分

$$f'(x_0) \approx \frac{-11f(x_0)+18f(x_1)-9f(x_2)+2f(x_3)}{6h}$$

第七章非线性方程

二分法

$$|x^*-x_n| \le \frac{b-a}{2^n}$$

不动点迭代

$$x_{n+1} = \varphi(x_n)$$

$$\varphi([a,b]) \subset [a,b], \qquad |\varphi'(x)| \le q < 1$$

$$|x^*-x_n| \le \frac{q}{1-q}|x_n-x_{n-1}|$$

Newton 迭代

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

单根：二阶收敛，　　重根：通常一阶收敛

$$x_{n+1} = x_n - m\frac{f(x_n)}{f'(x_n)}$$

收敛阶

$$\varphi'(x^*) = \cdots = \varphi^{(p-1)}(x^*) = 0, \qquad \varphi^{(p)}(x^*) \ne 0 \quad \Rightarrow \quad p \text{ 阶收敛}$$

第九章常微分方程数值解

Euler 类公式

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$$\begin{cases} \bar{y}_{n+1} = y_n + hf(x_n, y_n), \\ y_{n+1} = y_n + \dfrac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, \bar{y}_{n+1})\right] \end{cases}$$

$$\begin{cases} y_{n+\frac{1}{2}} = y_n + \dfrac{h}{2}f(x_n, y_n), \\ y_{n+1} = y_n + hf\left(x_n + \dfrac{h}{2}, y_{n+\frac{1}{2}}\right) \end{cases}$$

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right]$$

经典四阶 RK

$$\begin{cases} k_1 = f(x_n, y_n), \\ k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \\ k_3 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \\ k_4 = f(x_n + h, y_n + hk_3), \\ y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \end{cases}$$

高阶方程化一阶系统

$$y_1 = y, \quad y_2 = y', \quad y_3 = y'', \ldots$$

方法阶数

$$\text{局部截断误差} O(h^{p+1}) \quad \Rightarrow \quad \text{方法为} p \text{ 阶}$$

四步 Adams-Bashforth

$$y_{n+1} = y_n + \frac{h}{24}(55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3})$$

绝对稳定性

$$y' = \lambda y, \qquad z = h\lambda$$

$$y_{n+1} = R(z)y_n \quad \Rightarrow \quad \boxed{|R(z)| \le 1}$$

$$R_{\text{RK4}}(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

$$R_{\text{IE}}(z) = R_{\text{EM}}(z) = 1 + z + \frac{z^2}{2}$$

$$R_{\text{IM}}(z) = \frac{1 + \frac{z}{2}}{1 - \frac{z}{2}}$$

第十章偏微分方程差分

热方程显式格式

$$u_t = au_{xx}$$

$$r = \frac{\tau}{h^2}, \qquad u_i^{k+1} = ar\,u_{i-1}^k + (1 - 2ar)u_i^k + ar\,u_{i+1}^k, \qquad ar \le \frac{1}{2}$$

$$r = \frac{a\tau}{h^2}, \qquad u_i^{k+1} = r\,u_{i-1}^k + (1 - 2r)u_i^k + r\,u_{i+1}^k, \qquad r \le \frac{1}{2}$$

边界与初值

$$u(0,t) = u(1,t) = 0 \quad \Rightarrow \quad u_0^k = u_N^k = 0$$

$$u(x,0) = g(x) \quad \Rightarrow \quad u_i^0 = g(x_i)$$

一阶双曲方程

$$u_t + au_x = 0$$

$$\boxed{\left|\frac{ak}{h}\right| \le 1}$$

$$a = 1 \quad \Rightarrow \quad \boxed{\frac{k}{h} \le 1}$$

必背公式

$$e_r < \frac{1}{2a_1}10^{1-n}$$

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\prod(x - x_i)$$

$$A^TAa = A^Ty$$

$$S_n = \frac{h}{3}[f_0 + f_n + 4\sum f_{奇} + 2\sum f_{偶}]$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$|R(z)| \leq 1$$

$$r \leq \frac{1}{2} \quad \text{或} \quad ar \leq \frac{1}{2}$$